(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

VI Semester, April, 2024

Paper Code: EEE-324T **Subject: Introduction to Data Communication and Networking**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Define networking. What are the different components of computer network?	2	CO1
(b)	What are the differences with star and ring topology?	2	CO1
(c)	Differentiate between twisted pair cable and coaxial cable.	2	CO1
(d)	Differentiate between a bridge, hub and a switch.	2	CO2
(e)	What is HDLC?	2	CO2
2 (a)	Draw the OSI network architecture and explain the functionalities of every layer in detail.	5	CO1
(b)	Briefly explain the different types of network topologies. Write each of its advantages and disadvantages.	5	CO1
3 (a)	Explain the Shielded twisted pair (STP) and Unshielded twisted pair (UTP).	5	CO1
(b)	Explain the major component of a packet switch and their functions.	5	CO1
4 (a)	Explain the physical properties of Ethernet 802.3 with necessary diagram of Ethernet transceiver and adaptor.	5	CO2
(b)	Examine the various issues in the Data link layer.	5	CO2